2. A ferromagnetic core with a relative permeability of 2000 is shown in Figure 2. The dimensions are as shown in the diagram, and the depth of the core is 7 cm. The air gaps on the left and right sides of the core are 0.05 and 0.07 cm, respectively. If there are 300 turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux of the center the core? What is the flux density?

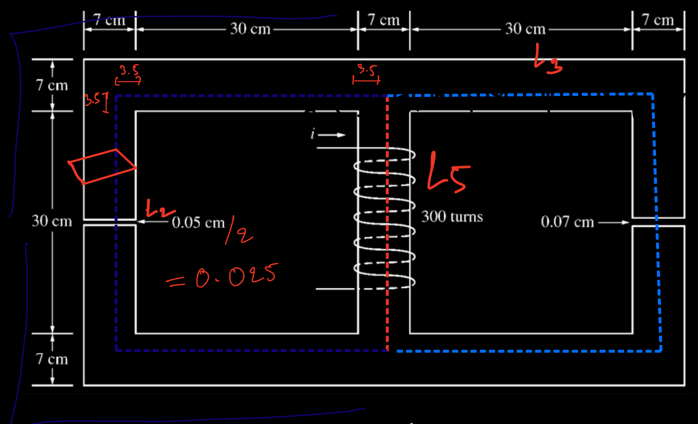
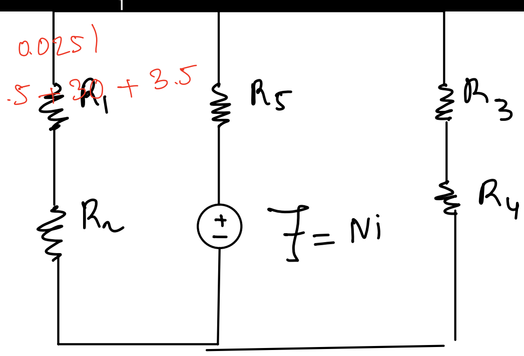


Figure 2

1

5



$$\frac{1}{R'} = \frac{1}{R_1+R_2} + \frac{1}{R_3+R_4}$$

$$R' = \frac{(R_1+R_2)(R_3+R_4)}{R_1+R_2+R_3+R_4}$$

$$\Rightarrow R = R_5 + R'$$

$$L_5 = 30 + 3.5 \times 2$$

$$L_2 = 0.05 \text{ cm} = 0.0005 \text{ m}$$

$$L_3 = 1.11 \text{ m} - 0.0007 = 1.1093$$

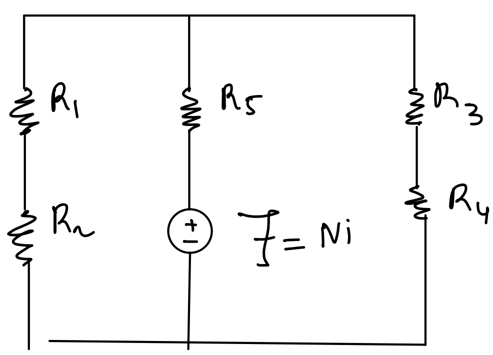
$$L_5 = 37 \text{ cm} = 37 \times 10^{-2} \text{ m} = 0.37$$

$$L_4 = 0.07 \text{ cm} = 0.0007 \text{ m}$$

$$L_1 = 1.11 - 0.0005 = 1.1095$$

- The equivalent reluctance of a number of reluctances in series is:

$$R_{total} = R_5 + \frac{(R_1+R_2)(R_3+R_4)}{R_1+R_2+R_3+R_4}$$



$$\frac{1}{R'} = \frac{1}{R_1+R_2} + \frac{1}{R_3+R_4}$$

$$R' = \frac{(R_1+R_2)(R_3+R_4)}{R_1+R_2+R_3+R_4}$$

$$\Rightarrow R = R_5 + R'$$

find reluctance

$$R_1 = \frac{l_1}{\mu_0 \mu_r \times A_1} = \frac{1.1093}{2\times10^3 \times 4\pi\times10^{-7} \times (7\times7)\times10^{-4}}$$

$\mu_r = 2\times10^3$

$R_1 = 90.07$ KAt/wb

$$R_2 = \frac{0.0005}{4\pi\times10^{-7}\times(7\times7)\times10^{-4}} = 81.201 \text{ KAt/wb}$$

$\mu_r = 1$

$$R_3 = \frac{1.105}{2\times10^3\times4\pi\times10^{-7}\times(49\times10^{-4})} = 90.093 \text{ KAt/wb}$$

$$R_4 = \frac{0.0007}{4\pi\times10^{-7}\times(7\times7)\times10^{-4}} = 113.68 \text{ KAt/wb}$$

$$R_5 = \frac{0.37}{4\pi\times10^{-7}\times(7\times7)\times10^{-4}\times2\times10^3}$$

$= 30.04$ KAt/wb

$$\Rightarrow R = 30.04 + \frac{(90.07+81.201)(90.093+30.04)}{90.07+81.201+90.093+30.04}$$

$R = 100.647$

flux center

$$\Phi_{center} = \Phi_{total} = \frac{\mathcal{F}}{R_{total}} = \frac{Ni}{R_{total}}$$

$$\Phi_{tot} = \frac{300 \times 1}{100.647} = 2.98$$

flux density,

$$B = \frac{\Phi_{tot}}{A} = \frac{2.98}{7 \times 7 \times 10^{-4}} = \boxed{608 T}$$